INTERACTIVE LEARNING TOOL FOR DYSLEXIA KIDS



RP26-IT-130

SLIIT
FACULTY OF COMPUTING

WHAT IS DYSLEXIA ?

Dyslexia is a relatively common, lifelong learning difficulty affecting up to 1 in 10 people. It mainly causes problems with reading, writing and spelling. It does not impact intelligence.



MAIN WEAKNESSES

Poor Working memory

Phonological awareness

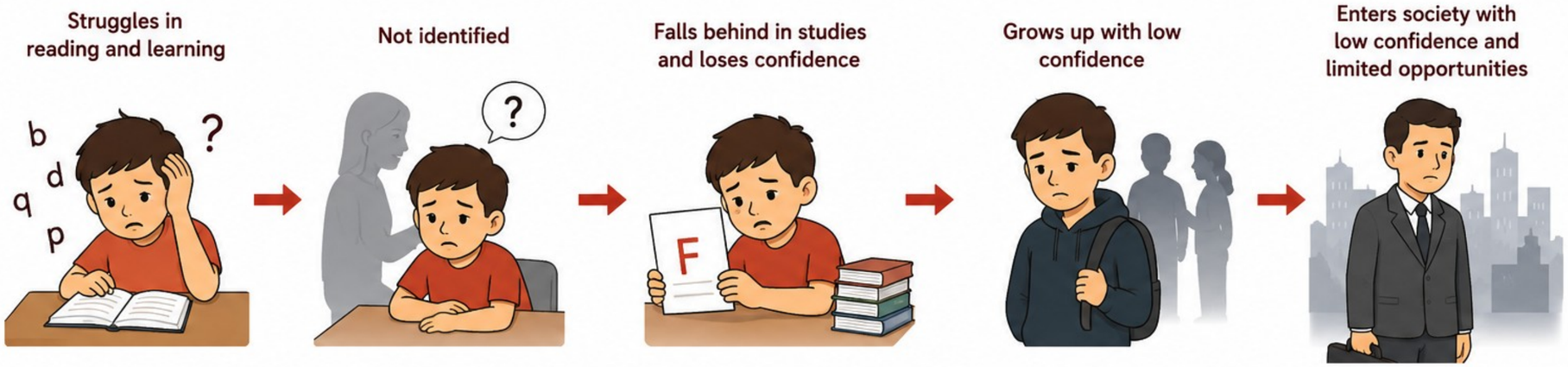


Reading Difficulty

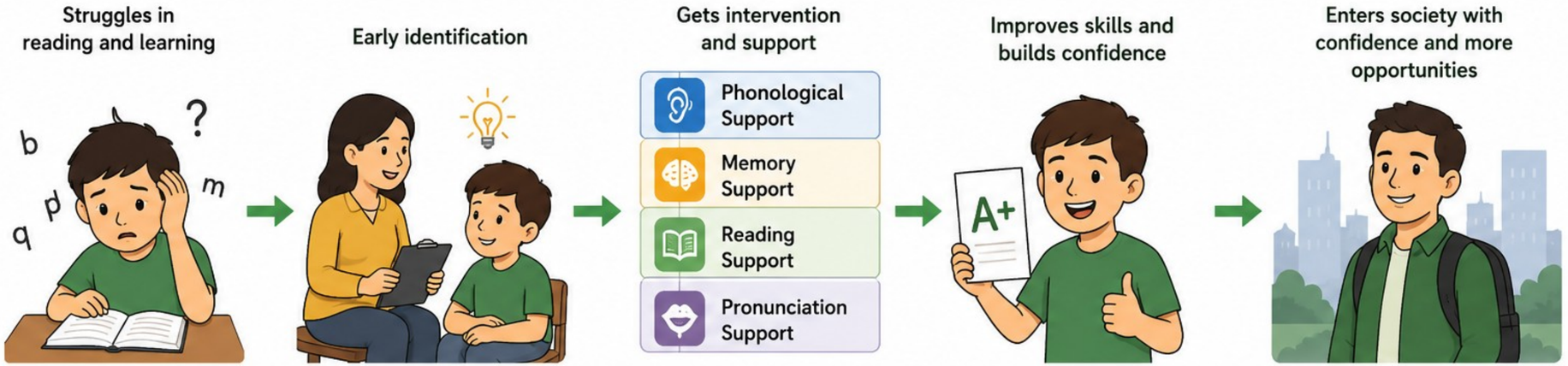
Pronounciation Diffyculty

IMPORTANCE OF EARLY IDENTIFICATION

WITHOUT IDENTIFYING DYSLEXIA



WITH EARLY IDENTIFICATION AND INTERVENTION SUPPORT



TARGET GROUPS



Students



Teachers



Parents

ANY EXISTING SOLUTIONS ?

- None of them address all 4 weakness together in a one single platform
- None of them focus on flexibilty for grade 2,3,4 students (Specially based on Local Syllabus)
- Language flexibility between Sinhala and English language (English for international school students)

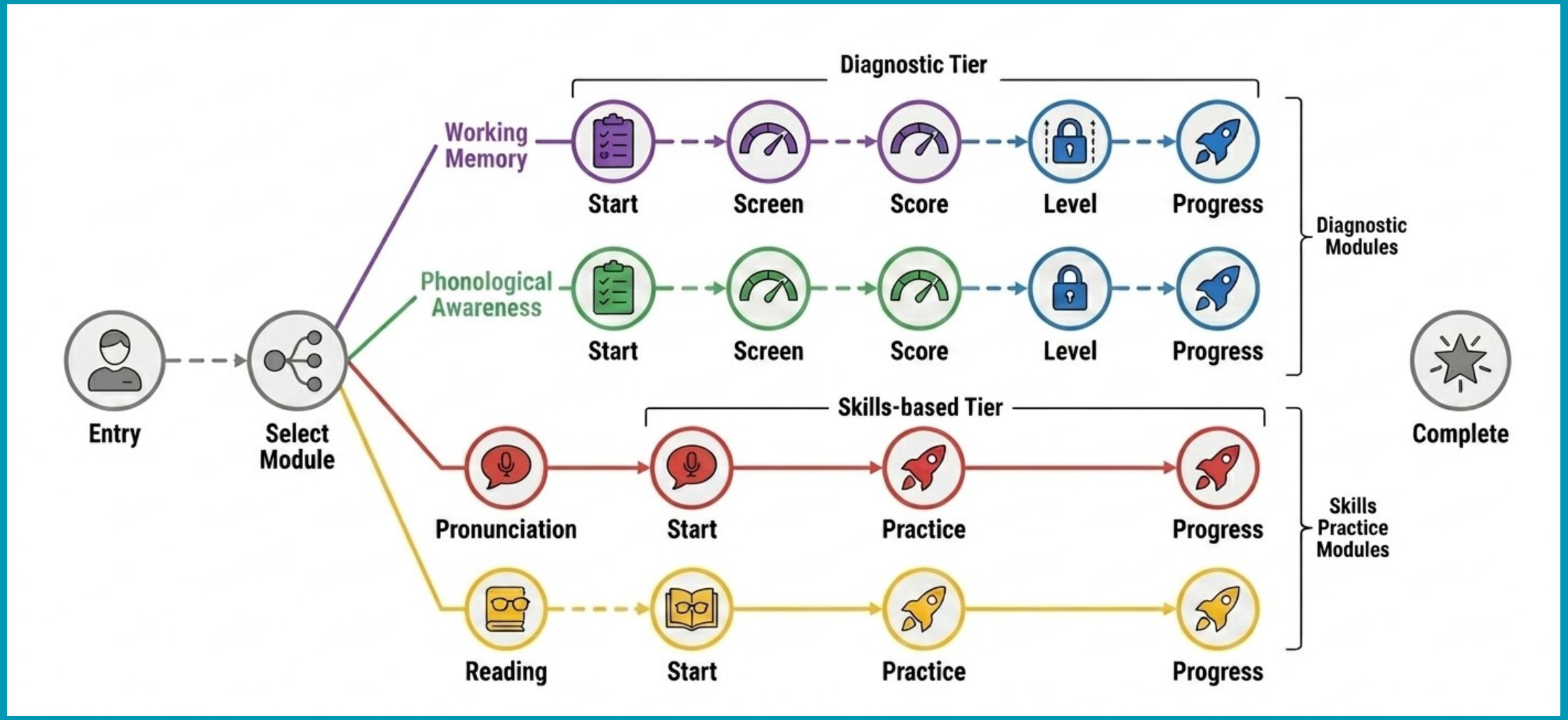
SOLUTION


LexiLand
Jungle Adventure
Dive in and start your
learning adventure!


LexiLand
කැලයේ අසිරිය විඳීමු
එන්න අපි කැලය ඇතුළට යමු


Let's Explore the Jungle!
1
WORKING MEMORY
10 Stars | Level 2
2
PHONOLOGICAL AWARENESS
15 Stars | Level 3
3
READING PROCESSING
20 Stars | Level 4
4
SPEECH PROCESSING
25 Stars | Level 5

SYSTEM DIAGRAM



WORKING MEMORY SEQUENCE MODULE

IT22127150

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Information Technology



MAIN AREAS OF WM

Pattern Recognition

Following Instructions

Brains Ability to process and hold the information for a short time of period



Word Memory

Number Memory

Working memory & sequencing weakness

Pattern recognition



▲ ● ▲ ● ▲ ■

Spots the rule fast

A8A8 → easy ✓

"Triangle, circle, triangle, circle..."

★★★

Word memory

Typical student



Heard: cat, tree, sun, bell, dog

cat | tree | sun | bell | dog

"cat, tree, sun, bell, dog!"

★★★

Dyslexic student



Heard: cat, tree, sun, bell, dog

sun | cat | ...

"I forgot the rest of them..."

☆☆☆

Following instructions

Typical student



1. Touch nose ✓
2. Clap hands ✓
3. Sit down ✓

3/3 steps done ✓

★★★

Dyslexic student



1. Touch nose ~
2. What's next? X
3. Forgotten

1/3 steps done X

☆☆☆

Number memory

Typical student



Repeat: 7-3-9-1-5

7 3 9 1 5

✓ ✓ ✓ ✓ ✓

5/5 digts correct ✓

★★★

Dyslexic student



Repeat: 7-3-9-1-5

3 7 _ _

X X – –

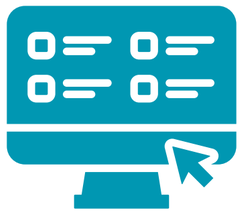
2/5 transposed X

☆☆☆

Typical: strong memory

Dyslexic: memory weakness

HOW I ADDRESED THE PROBLEM



Screening Test

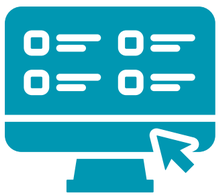


Categorize



Improve

SCREENING TEST QUESTIONNAIRE



Screening Test

Consist of 15 Questions

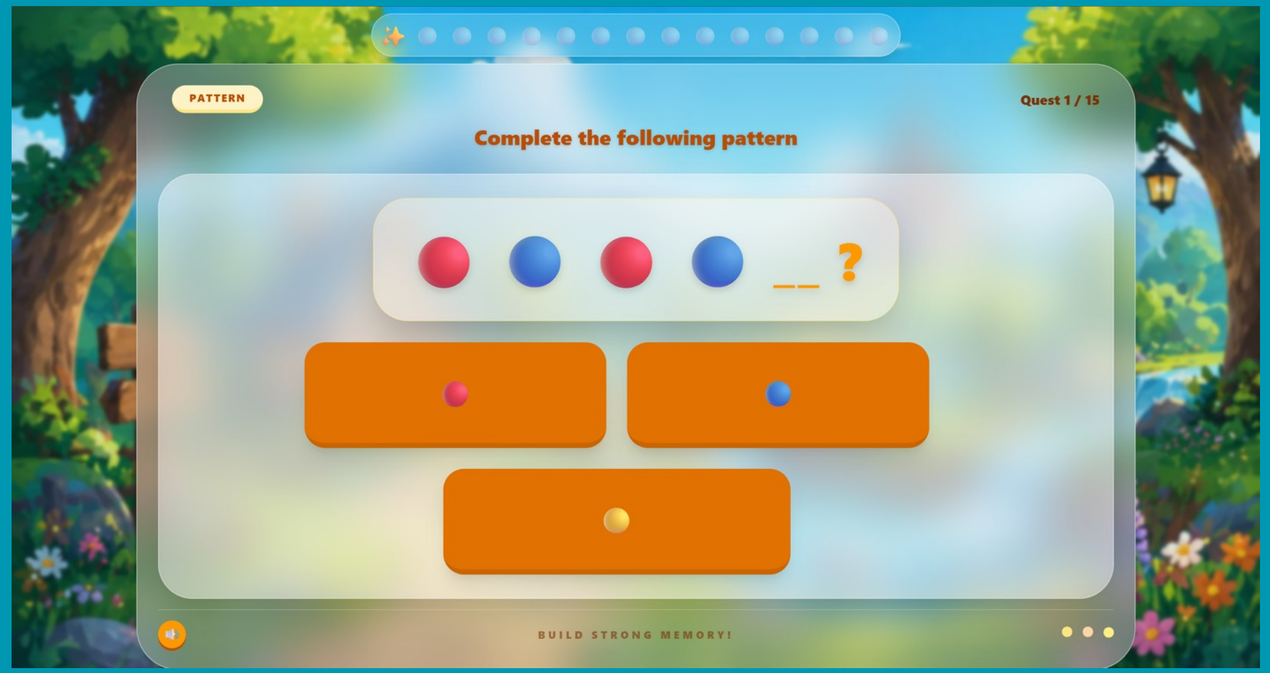
4 Qs - Pattern Recognition

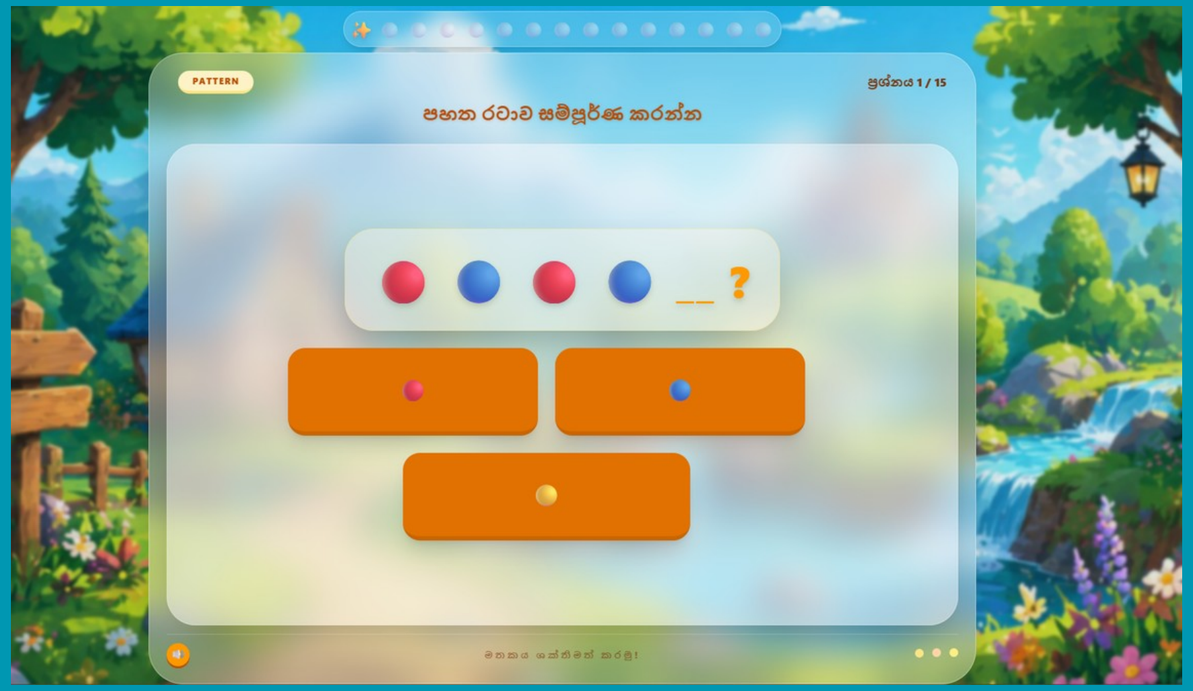
3 Qs - Word Memory

4 Qs - Following Instructions

4 Qs - Number Memory

SAMPLE INTERFACES OF SCREENTEST





CATERGORIZING



Catergory

Well Performing

Average

Low Performing

(Who might have the risk of having Dyslexia)

USER FEEDBACK ON PROTOTYPES

- There were some Drag and Drop options - Multiple Choice Answers
- Teacher were evaluate our questions and asked us to simplify the questionnair
- Students were exicited to particiapate for the activities

MACHINE LEARNING MODEL

```
[ ]
# ==============================
# 8. Train Random Forest Model
# ==============================
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42,
    class_weight="balanced"
)

model.fit(X_train, y_train)

print("Model Trained Successfully")
print("Classes:", model.classes_)
```

```
Model Trained Successfully
Classes: ['Average' 'High' 'Low']
```

```
[ ]
# ==============================
# 13. Cross Validation
# ==============================
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_scores = cross_val_score(model, X, y, cv=cv, scoring="accuracy")

print("CV Scores:", cv_scores)
print("Mean Accuracy:", cv_scores.mean())
print("Std Dev:", cv_scores.std())
```

```
CV Scores: [0.88888889 0.77777778 1.         0.66666667 0.75      ]
Mean Accuracy: 0.8166666666666667
Std Dev: 0.11600340565456167
```

PHONOLOGICAL AWARENESS MODULE

IT22141088

Fernando K. H. D. M.

Information Technology



WHAT IS PHONOLOGICAL AWARENESS ?

Phonological Awareness Weakness

Visual Processing Weakness



Literacy Skills

i. Orthographic Skills Weakness

ii. Segmentation & Counting Weakness

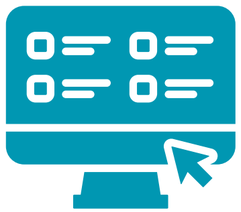
PHONOLOGICAL AWARENESS

	1. Phonological Awareness	2. Visual Processing	3. Orthographic Skills	4. Segmentation & Counting
Dyslexic Student	"ගස" යන වචනයේ මුල් අකුර කුමක්ද?  ය ✗ / ග ✗ / හ ✓ මුල් ගබදය හඳුනා ගැනීමට අපහසුයි.	වෙනස් අකුර තෝරන්න  ග ✗ / ග ✗ / හ ✓ අකුරු අතර වෙනස්කම් දැකීටට අපහසුයි.	"____ල" හිස්තැන පුරවන්න  ම ✗ / බ ✓ / න ✗ නිවැරදි අකුර තෝර ගැනීමට අපහසුයි.	"කබය" යන වචනයේ අකුරු කියක් ඇත්ද?  2 ✓ / 3 ✗ / 4 ✗ අකුරු ගණන් කිරීමට අපහසුයි.
Normal Student	"ගස" යන වචනයේ මුල් අකුර කුමක්ද?  ය ✗ / ග ✓ / හ ✗ මුල් ගබදය නිවැරදිව හඳුනා ගනී.	වෙනස් අකුර තෝරන්න  ග ✗ / ග ✗ / හ ✓ අකුරු අතර වෙනස්කම නිවැරදිව හඳුනා ගනී.	"____ල" හිස්තැන පුරවන්න  ම ✓ / බ ✗ / න ✗ නිවැරදි අකුර තෝරා ලියයි.	"කබය" යන වචනයේ අකුරු කියක් ඇත්ද?  2 ✗ / 3 ✓ / 4 ✗ අකුරු ගණන් නිවැරදිව කරයි.

PHONOLOGICAL AWARENESS

	1. Phonological Awareness	2. Visual Processing	3. Orthographic Skills	4. Segmentation & Counting
Dyslexic Student	What is the last letter of "shop"?  p ○ q ○ b ✗ Struggles to hear and identify the last sound in the word.	Find the different letter.  b ○ b ○ d ✗ Has difficulty noticing visual differences between letters.	Which is the correct spelling? (image: dog)  dog ✗ qog ○ pog ○ Struggles to recognize the correct spelling.	How many letters are in the word "Park"?  4 ○ 3 ○ 2 ✗ Finds it hard to count the letters in the word.
Normal Student	What is the last letter of "shop"?  p ○ q ○ b ✓ Easily identifies the last sound and letter.	Find the different letter.  b ○ b ○ d ✓ Easily identifies the different letter.	Which is the correct spelling? (image: dog)  dog ✓ qog ○ pog ○ Recognizes and selects the correct spelling.	How many letters are in the word "Park"?  4 ✓ 3 ○ 2 ○ Counts the letters correctly.

HOW I ADDRESED THE PROBLEM



Screening Test



Categorize



Improve

HOW I ADDRESED THE PROBLEM

Screening Test Consist of 15 Questions

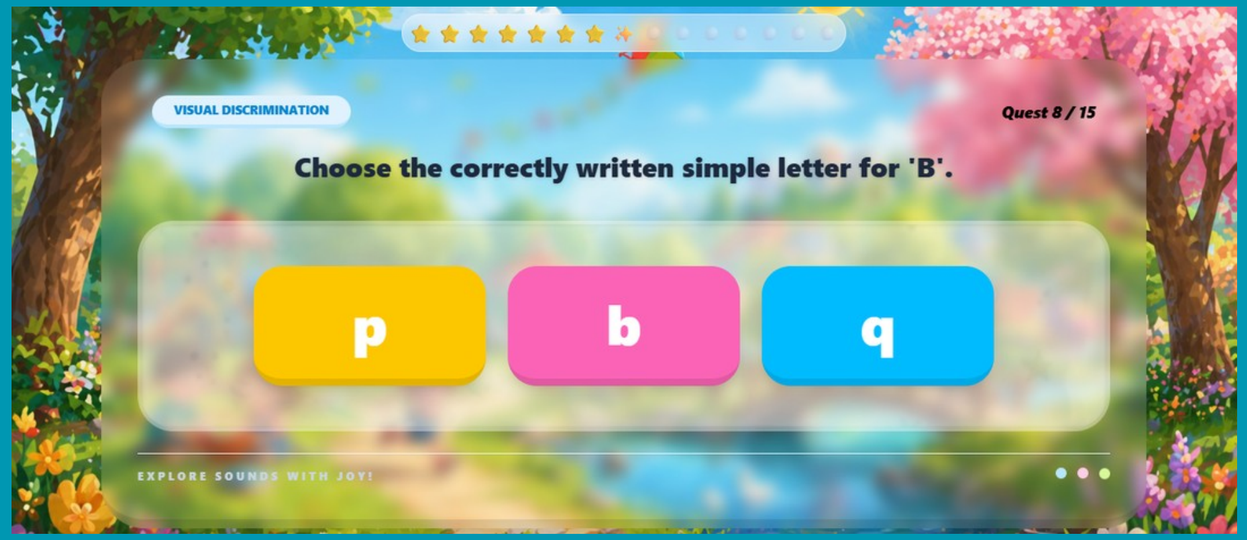
5 Qs - Phonological Awareness

6 Qs - Visual Processing

4 Qs - Literacy Skills

SAMPLE INTERFACES OF SCREENTEST





CATERGORIZING



Categorize

01. High Risk Students

(Students who might have the risk of having Dyslexia)

02. Average Students

03. Well Performing Students

USER FEEDBACK ON PROTOTYPE

- Drag-and-drop activities were replaced with MCQs since the students were more familiar with simple question formats.
- The questions were finalized with our external supervisor and later simplified based on teacher feedback to make them easier for children to understand.
- Students were excited and actively participated in the activities.

MACHINE LEARNING MODEL

Create and train the Random Forest Model

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=100,
    max_depth=5,
    random_state=42
)

model.fit(X_train, y_train)

print("Model trained successfully.")
```

```
Model trained successfully.
```

Cross Validation

```
from sklearn.model_selection import cross_val_score

cv_scores = cross_val_score(
    model, X, y,
    cv=5,
    scoring="accuracy"
)

print("Cross-validation scores:", cv_scores)
print("Mean CV accuracy:", cv_scores.mean())
print("Std CV accuracy:", cv_scores.std())
```

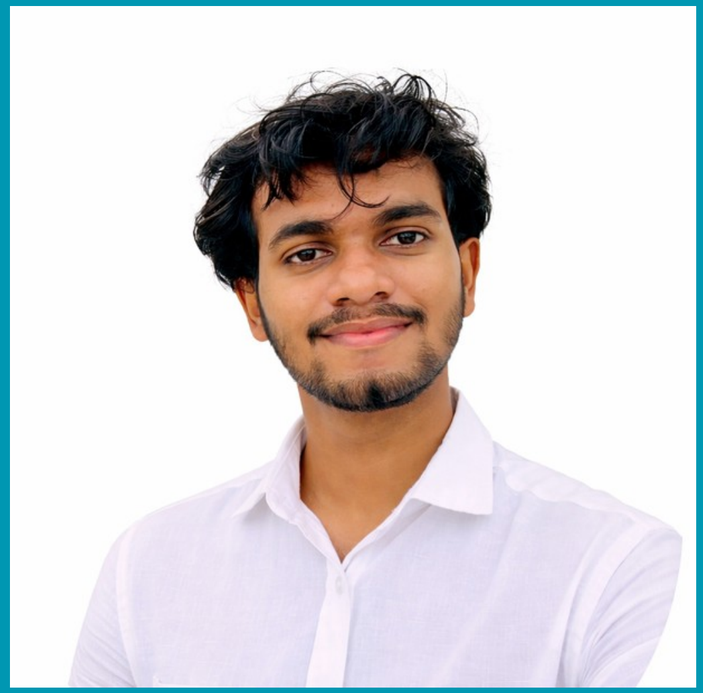
```
Cross-validation scores: [0.9        0.9        0.9        1.         0.88888889]
Mean CV accuracy: 0.9177777777777777
Std CV accuracy: 0.0413357227505295
```

PRONUNCIATION & SPEECH SOUND LEARNING MODULE

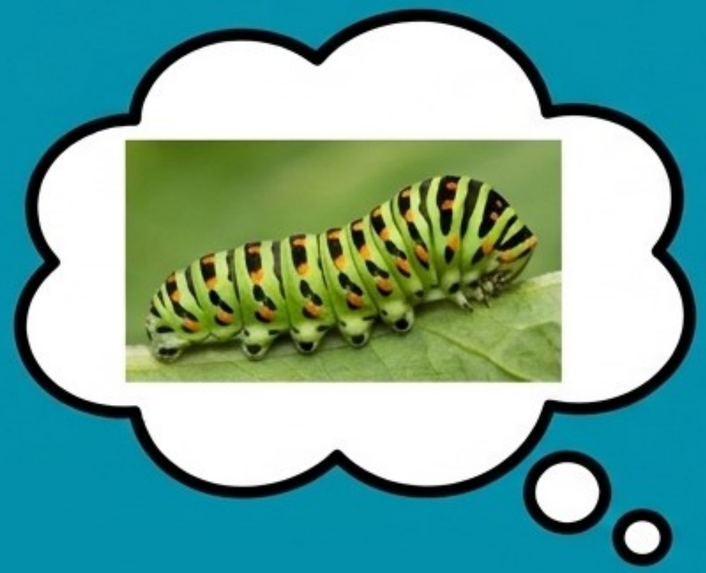
IT22027542

Navod J.N

Information Technology



THE PHONOLOGICAL OUTPUT GAP



C-A-T-E-R-P-L-L-A-R

Visual Input

Speech Production

micro-jumble

Output A: (Jumbled Phonemes)

"Cat-a-piller"

Jumbled Sounds

micro-jumble

Output B: (Spoonerism)

"Pater-cillar"

Sound Swapping

Output C: (Omission)

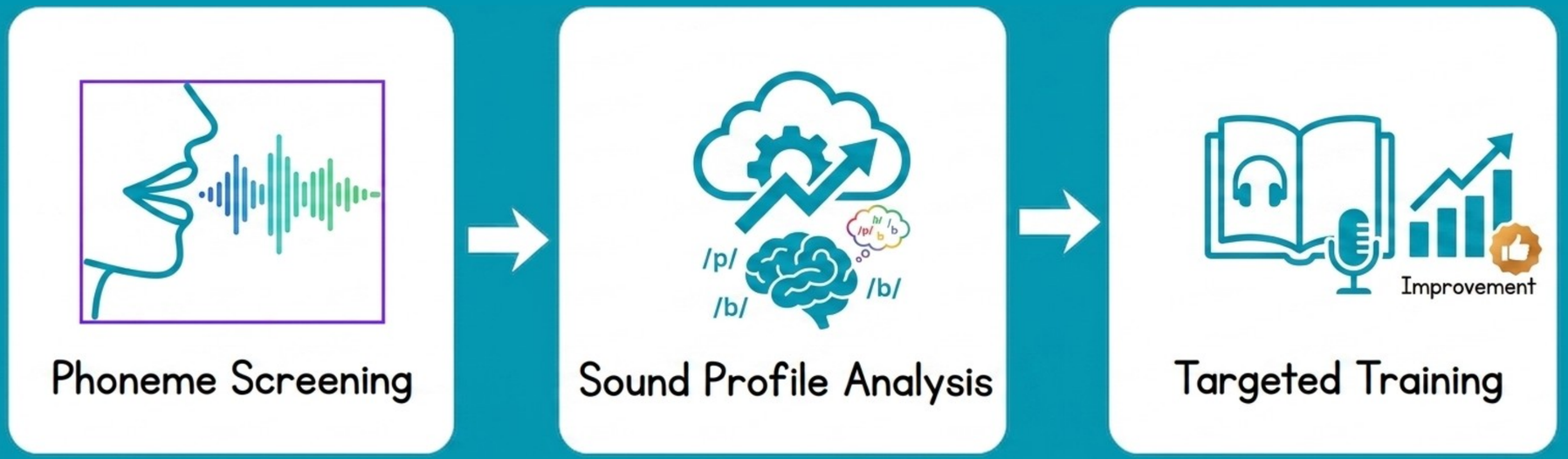
"Cater-pi-la"

Syllable Omission

Visual Input

Speech Production

HOW I ADDRESED THE PROBLEM



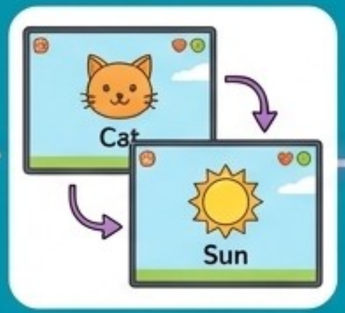
ADAPTIVE SPEECH REMEDIATION WORKFLOW

START



START

1. BASELINE SCREENING



Grade 2-4 Content

2. PHONETIC ANALYSIS



Uses Web Speech NLP / Error Clustering Algorithm

IF PERFORMANCE?

3A. ADAPTIVE REMEDIATION

If low performance (below 50%)

e.g. C-A-T

Syllable Segmentation Activities

Scaffolding Support

3B. PROGRESSION

If high performance (above 80%)

e.g. CATERPILLAR

Complex Multi-syllabic Words

Speed Challenges

4. RE-ASSESSMENT & GAME DATA

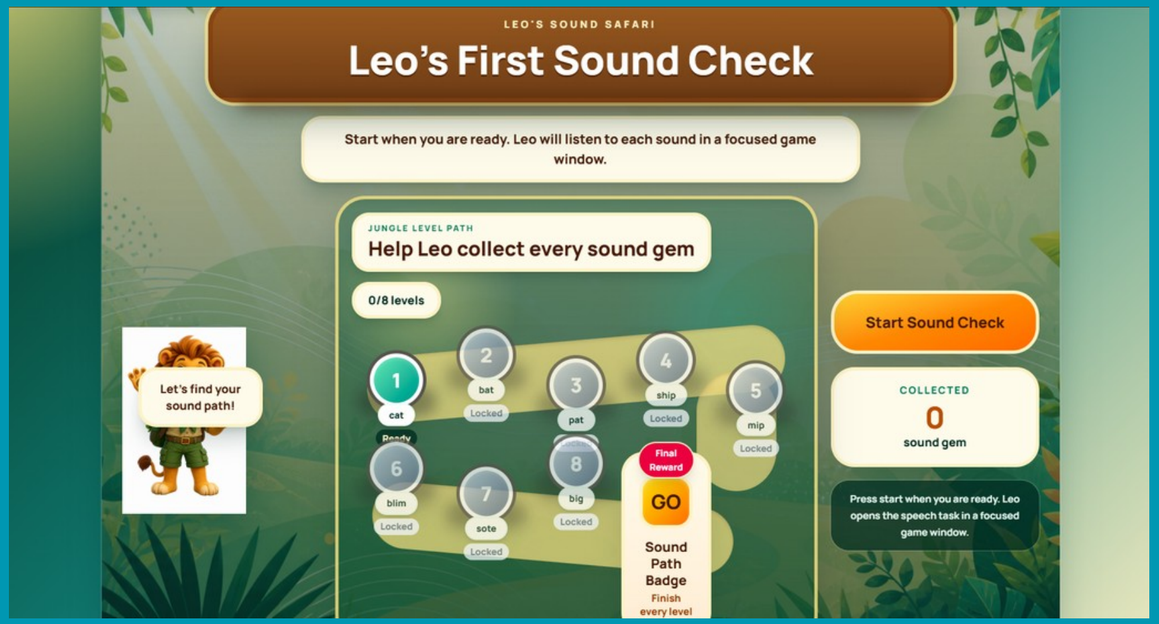


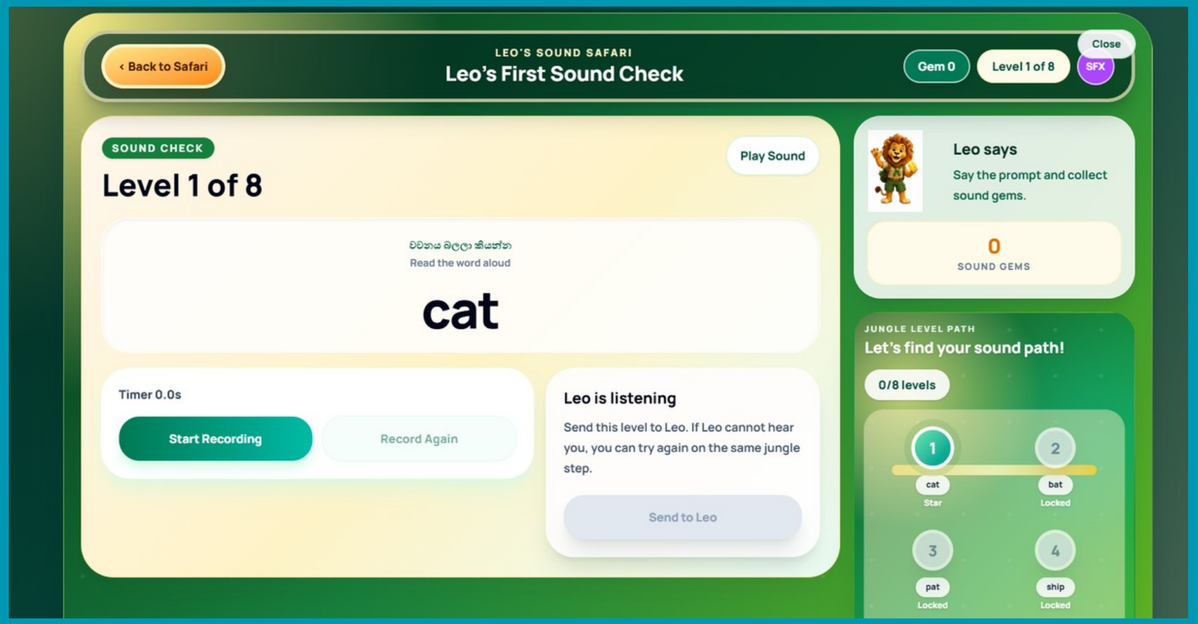
4. RE-ASSESSMENT & GAME DATA



Teacher/Admin Performance Metrics

USER INTERFACES OF ACTIVITY





CATERGORIZING



Categorize

01. High Risk Students

(Students who might have the risk of having Dyslexia)

02. Average Students

03. Well Performing Students

MACHINE LEARNING MODEL

```
Training RandomForest...
Training GradientBoosting...

==============================
MODEL: random_forest
Accuracy: 0.678125
Macro F1: 0.5903360508703557
                precision    recall  f1-score   support

  high_support       0.59      0.86      0.70        99
   low_support       0.73      0.88      0.79       359
medium_support       0.56      0.19      0.28       182

      accuracy                           0.68       640
     macro avg       0.62      0.64      0.59       640
  weighted avg       0.66      0.68      0.63       640

Confusion matrix:
[[ 85  10   4]
 [ 21 315  23]
 [ 39 109  34]]
```

EYE-TRACKING BASED READING MODULE

IT22266064

KHAN N.A.B

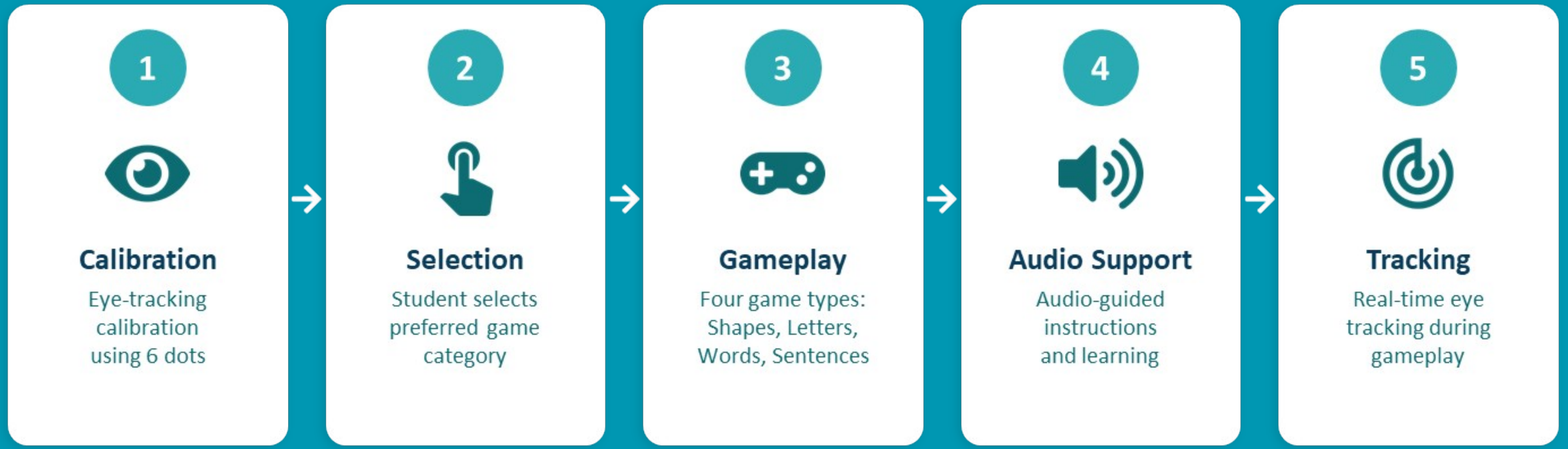
Information Technology



PROBLEM OF MY COMPONENT

- Difficulty reading letters, words, and sentences
- Losing focus and concentration while reading
- Skipping or rereading words and lines
- Difficulty tracking text with eye movements
- Lack of personalized learning support

HOW I ADDRESED THE PROBLEM



HOW I ADDRESED THE PROBLEM

Interactive Learning Games

Engaging shapes, letters, words & sentence games

Real-Time Eye Tracking

Monitor gaze patterns during reading activities

Audio Assistance

Voice-guided instructions for comprehension

Personalized Learning

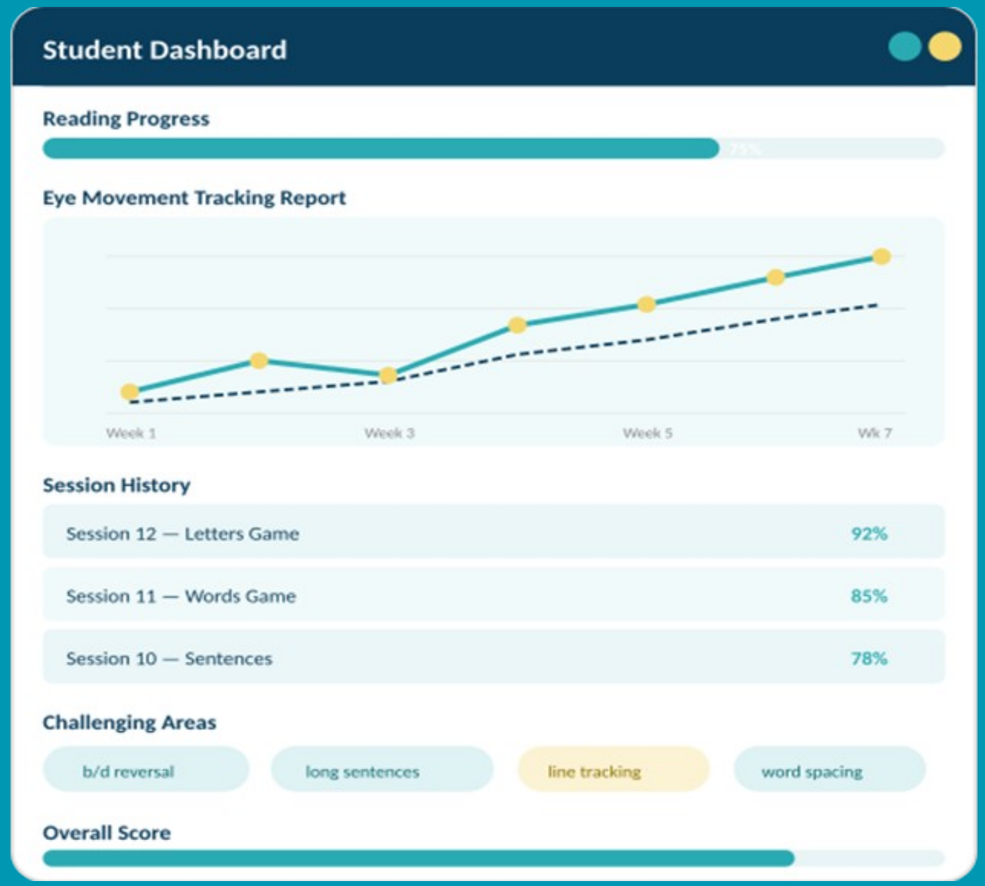
Adaptive difficulty based on performance

Performance Analysis

Detailed reports on reading progress

Dashboard Reports

Visual summaries for teachers & parents



FINAL OUTCOME AND IMPACT

Improved Reading Focus

Eye tracking helps students maintain focus and engagement during reading

Pattern Identification

Identify challenging reading patterns and areas needing targeted support

Personalized Learning

Adaptive learning paths tailored to each student's unique needs

Progress Tracking

Continuous monitoring of student improvement with visual reports

Teacher & Parent Insights

Actionable reports to help educators and parents support students



COMMERCIALIZATION PLAN



THANK YOU !!

APPENDIXES

APPENDIXES





